

## CE 310 — Week 9 Weekly Quiz

**Week:** 9 | **Topic:** Simple Linear Regression — Concepts and Python/scipy

**Due:** Closes **8:00 AM Tuesday, Oct 27** — the day before Exercise 9 and Homework 9 are due. Taken on D2L

**Points:** 15 questions × 1 pt each = 15 pts

**Attempts:** Two — the higher score counts

**Time limit:** 25 minutes

**Instructions:** Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

**This quiz covers the Monday lecture and the Wednesday Python tutorial** — simple linear regression, the role of  $R^2$ , residuals, and the Python tools that go with them. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

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### Questions

**Q1.** Simple linear regression gives you an equation of the form  $\hat{y} = \beta_0 + \beta_1 \cdot x$ . What does this equation allow you to do that correlation alone does **not**?

- A) Measure how strong the linear relationship is between two variables
  - B) Test whether two variables are statistically independent
  - C) Determine whether the relationship between  $x$  and  $y$  is causal
  - D) Predict a specific numerical value of  $y$  for any given value of  $x$
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**Q2.** In the regression equation  $\hat{y} = \beta_0 + \beta_1 \cdot x$ , the slope  $\beta_1 = 1.16 \text{ kWh}/^\circ\text{F}$  for a building's cooling model. Which statement correctly interprets this slope?

- A) For every  $1^\circ\text{F}$  increase in outdoor temperature, predicted cooling demand increases by  $1.16 \text{ kWh/hr}$
  - B) When outdoor temperature is  $0^\circ\text{F}$ , the building uses  $1.16 \text{ kWh}$  of cooling
  - C) Cooling demand increases  $1.16$  times faster than outdoor temperature
  - D) The building uses  $1.16 \text{ kWh}$  of cooling on a typical day, averaged across the whole year of the record
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**Q3.**  $R^2 = 0.60$  for a regression model predicting cooling energy from outdoor temperature. Which statement is the most accurate engineering interpretation?

- A) 60% of outdoor temperature variation is explained by cooling energy
  - B) Outdoor temperature explains 60% of the variance in cooling energy
  - C) The regression line passes through 60% of the data points
  - D) The model is 60% accurate and should not be used for design decisions
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**Q4.** [Select all that apply] A regression predicting cooling energy from outdoor temperature returns  $R^2 = 0.60$ . Which of the following are true?

- (A) 60% of the variance in cooling energy is linearly explained by outdoor temperature
  - (B) The remaining 40% is variation the model does not account for — occupancy, equipment loads, time of day
  - (C) The model's predictions are correct 60% of the time
  - (D)  $R^2$  here is the square of the Pearson  $r$  between the two variables
  - (E) A higher  $R^2$  would establish that outdoor temperature causes cooling demand
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**Q5.** [True/False] A fitted linear regression can predict a negative cooling load at low outdoor temperatures, even though the chiller can only ever be off or running — so the fitted line has a range over which it should not be used.

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**Q6.** A **residual** is defined as:

- A) The slope of the regression line at a given data point
  - B) The square root of  $R^2$ , always expressed on the same scale as the outcome variable
  - C) The difference between the observed  $y$ -value and the model's predicted value
  - D) The standard deviation of the predictor variable  $x$
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**Q7.** A colleague says “the regression line and the correlation coefficient tell you the same thing.” Which response is correct?

- A) True — correlation  $r$  and the regression slope  $\beta_1$  are identical
- B) True —  $R^2 = r$ , so they convey the same information
- C) False — correlation measures strength and direction; regression gives the predictive equation
- D) False — correlation applies only to Normal data; regression applies only to skewed data

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**Q8.** Your Week 9 model predicts cooling energy from outdoor temperature:  $\hat{y} = \beta_0 + \beta_1 \cdot x$ , with  $\beta_1 = 1.16 \text{ kWh/}^\circ\text{F}$ . A data scientist describes the identical model in machine-learning terms. Which description is correct?

- A) Outdoor temperature is the label, cooling energy is the feature, and  $R^2$  is the loss function
- B)  $\beta_1$  is the label, outdoor temperature is the weight, and the residual plot is the performance metric
- C) Both variables are features; the label is supplied separately once the model is deployed
- D) Cooling energy is the label, outdoor temperature is the feature, and least squares is the loss function being minimized

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**Q9.** [Matching] Match each Python expression to what it produces.

Choices:

- (A) `scipy.stats.linregress(x, y)`
- (B) `df.head()`
- (C) `df['A'].corr(df['B'])`
- (D) `df.describe()`
- (E) `y - y_pred`

Matches:

1. The Pearson  $r$  between two columns
2. Count, mean, standard deviation and quartiles for every numeric column
3. Slope, intercept,  $r$ ,  $p$ -value and standard error
4. The residuals — what the model missed at each observation

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**Q10.** [Ordering] Put the steps of fitting and checking a simple linear regression in the order you would carry them out.

- (A) Plot  $y$  against  $x$  and confirm the relationship looks broadly linear
- (B) Fit the line and read off the slope, intercept and  $r$
- (C) Plot the residuals against  $x$  to check the fit is equally good across the range
- (D) Predict from the equation, stating the slope's units alongside the number

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**Q11.** `slope, intercept, r_val, p_val, std_err = stats.linregress(x, y)`. You need  $R^2$  for the memo. What do you report?

- A) `r_val`, which is already  $R^2$
- B) `r_val**2`

- C) `abs(r_val)`
  - D) `1 - r_val`
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**Q12.** Your slope comes out around 0.86 where about 1.16 is expected. What is the most likely cause?

- A) x and y were passed the wrong way round, fitting the reverse regression
  - B) The data needs sorting by x before `linregress` will fit it correctly
  - C) `linregress` requires numpy arrays and silently mis-handles pandas Series
  - D) Missing values shifted one column against the other without any warning
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**Q13.** What does `p_val` from `linregress` actually test?

- A) Whether the model is a good fit to the data overall, across all points
  - B) Whether the slope differs from zero — whether any linear relation exists
  - C) Whether the residuals are Normally distributed around the fitted line
  - D) Whether the data contains outliers large enough to distort the fit
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**Q14.** You plot residuals against fitted values. What are you looking for?

- A) A shapeless band around zero; a curve or funnel means something is missing
  - B) A straight upward line, showing the model has captured the underlying trend
  - C) All residuals positive, confirming the model never under-predicts anywhere
  - D) Residuals reported in the same units as  $R^2$ , which makes them comparable
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**Q15.** You use the fitted model to predict cooling energy at 118 °F, when the hottest hour in the data is 112 °F. What is the concern?

- A) None — a fitted linear model stays valid across all real numbers
- B) It is extrapolation: nothing supports the relation holding past 112 °F
- C) `linregress` refuses inputs that fall outside the range it was fitted on
- D) The prediction will come out too low, because the model is strictly linear